

06.-Registro-de-Imágenes

Para esta actividad se publican y verifican las dos imágenes creadas en el componente anterior:

- **tarjeta-notas-v2:1.0**
- **to-do-list-v1:1.0**

Comenzaremos con Docker Hub como primer repositorio: se etiquetan y publican (push) ambas imágenes, se eliminan las versiones locales para forzar una descarga real, y luego se descargan (pull) nuevamente para crear los contenedores y confirmar que funcionan correctamente a partir de la imagen descargada. Una vez completada esta actividad en Docker Hub, se replica el mismo proceso en AWS ECR, como segundo repositorio.

Verificamos la sesión activa en Docker Hub

docker login

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker login
Authenticating with existing credentials... [Username: jezuz1904]

Info → To Login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
```

Confirmamos que las imágenes a publicar existen localmente:

docker images

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
apache/kafka:3.8.0	c89f315cff96	594MB	207MB	U
dhi.io/jenkins:2.580-debian-dev	fa9cf20c30e4	820MB	246MB	
dhi.io/prometheus:3.14	f18dc10d186c	267MB	52.1MB	
eclipse-mosquitto:2.0.18	d12c8f80dfc6	21.5MB	6.43MB	U
grafana/grafana:11.3.0	a0f881232a6f	646MB	133MB	U
mariadb:jammy	e101f9db3191	555MB	127MB	U
mysql:9.3.0	b9d8b7ec6e6a	1.19GB	273MB	
phpmyadmin:5.2.0-apache	7c8751488b72	736MB	180MB	
postgres:13	4689940c6838	624MB	161MB	
postgres:16	f1c3376c26f2	642MB	166MB	
postgres:latest	4ef4dbc939d6	650MB	168MB	
redis:7	71da9275c5f3	170MB	46MB	
tarjeta-notas-v2:1.0	7bce47c9d128	215MB	51.7MB	U
to-do-list-v1:1.0	06c0dd99463d	198MB	48.3MB	U
ubuntu:26.04	2260313b31c8	160MB	45.3MB	

Etiquetado de las imágenes

Docker Hub exige que toda imagen propia lleve como prefijo el nombre de usuario, seguido del nombre de la imagen. Por esta razón, se etiquetan ambas imágenes antes de publicarlas:

Tagamos las imágenes y verificamos si cumple con el requisito establecido por Docker Hub

```
docker tag tarjeta-notas-v2:1.0 jezuz1904/tarjeta-notas-v2:1.0
```

```
docker tag to-do-list-v1:1.0 jezuz1904/to-do-list-v1:1.0
```

```
docker images | grep jezuz1904
```

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker tag tarjeta-notas-v2:1.0 jezuz1904/tarjeta-notas-v2:1.0
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker tag to-do-list-v1:1.0 jezuz1904/to-do-list-v1:1.0
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker images | grep jezuz1904
jezuz1904/tarjeta-notas-v2:1.0    53e446d7fea4    215MB    51.7MB    U
jezuz1904/to-do-list-v1:1.0    06c0dd99463d    198MB    48.3MB    U
```

Publicación de las imágenes en Docker Hub:

```
docker push jezuz1904/tarjeta-notas-v2:1.0
```

```
docker push jezuz1904/to-do-list-v1:1.0
```

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker push jezuz1904/tarjeta-notas-v2:1.0
The push refers to repository [docker.io/jezuz1904/tarjeta-notas-v2]
6a9ac1617861: Pushed
0249bd96a5ad: Pushed
44136fa355b3: Pushed
4fae004c1543: Pushed
b2cbbfe903b0: Pushed
fff4e2c1b189: Pushed
8604042f67ad: Pushed
2b1c10b70b05: Pushed
794b77d4e2cc: Pushed
84fa716861fb: Pushed
49030ba06699: Pushed
1.0: digest: sha256:53e446d7fea4df5d62587ed97734a0ab496b3195764088d6f04cb87d4abce295 size: 856
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker push jezuz1904/to-do-list-v1:1.0
The push refers to repository [docker.io/jezuz1904/to-do-list-v1]
5c5309c50c96: Pushed
44136fa355b3: Mounted from jezuz1904/tarjeta-notas-v2
dbd0b7849e6c: Pushed
6310eb16bf42: Mounted from library/postgres
1560cfed01dc: Pushed
56235e424563: Pushed
8de16dab4666: Pushed
12e7896951c2: Pushed
9dca8d907a9a: Pushed
10cba69f7e13: Pushed
ab7715f487d9: Pushed
1.0: digest: sha256:06c0dd99463d252b282359f7e4da5da1f60c380b7823c15b0685a4aada1212b4 size: 856
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker info | grep Username
```

The screenshot shows the Docker Hub web interface. At the top, there's a navigation bar with 'Explore' and 'My Hub' tabs, a search bar, and utility icons. The left sidebar shows the user profile 'jezuz1904' and a menu with 'Repositories', 'Hardened Images', 'Collaborations', 'Settings', and 'Billing'. The main content area is titled 'Repositories' and shows 'All repositories within the jezuz1904 namespace.' Below this is a search bar and a table of repositories.

Name	Last Pushed	Contains	Visibility
jezuz1904/to-do-list-v1	about 1 hour ago	IMAGE	Public
jezuz1904/tarjeta-notas-v2	about 1 hour ago	IMAGE	Public

Descarga de las imágenes publicadas:

docker pull jezuz1904/tarjeta-notas-v2:1.0

docker pull jezuz1904/to-do-list-v1:1.0

```
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker pull jezuz1904/tarjeta-notas-v2:1.0
1.0: Pulling from jezuz1904/tarjeta-notas-v2
44136fa355b3: Already exists
8249bd96a5ad: Download complete
Digest: sha256:53e446d7fea4df5d62587ed97734a0ab496b3195764008d6f04cb87d4abce295
Status: Downloaded newer image for jezuz1904/tarjeta-notas-v2:1.0
docker.io/jezuz1904/tarjeta-notas-v2:1.0
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker pull jezuz1904/to-do-list-v1:1.0
1.0: Pulling from jezuz1904/to-do-list-v1
Digest: sha256:06c0dd99463d252b282359f7e4da5da1f60c380b7823c15b0685a4aada1212b4
Status: Downloaded newer image for jezuz1904/to-do-list-v1:1.0
docker.io/jezuz1904/to-do-list-v1:1.0
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
apache/kafka:3.8.0	c89f315cff96	594MB	207MB	U
dhi.io/jenkins:2.580-debian-dev	fa9cf20c30e4	820MB	246MB	
dhi.io/prometheus:3.14	f18dc10d186c	267MB	52.1MB	
eclipse-mosquitto:2.0.18	d12c8f80dfc6	21.5MB	6.43MB	U
grafana/grafana:11.3.0	a0f881232a6f	646MB	133MB	U
jezuz1904/tarjeta-notas-v2:1.0	53e446d7fea4	215MB	51.7MB	U
jezuz1904/to-do-list-v1:1.0	06c0dd99463d	198MB	48.3MB	U

Creación de contenedores a partir de las imágenes descargadas:

docker container run -d \

> -p 3000:3000 \

> --name DOCKER-HUB-TARJETA-NOTAS-V2 \

> jezuz1904/tarjeta-notas-v2:1.0

docker container run -d \

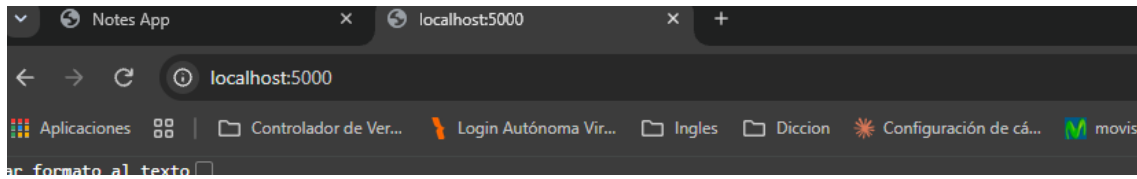
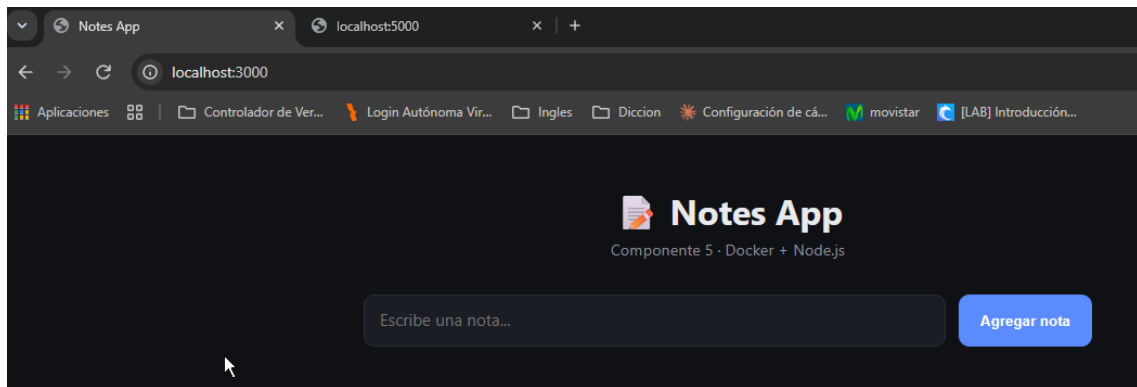
> -p 5000:5000 \

> --name DOCKER-HUB-TARJETA-NOTAS-V1 \

> jezuz1904/to-do-list-v1:1.0

```
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container run -d \
> -p 3000:3000 \
> --name DOCKER-HUB-TARJETA-NOTAS-V2 \
> jezuz1904/tarjeta-notas-v2:1.0
98460031f3b3cb9f1a1bf6af1cf9887df70038daf8a59974f596a91d5f664176
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container run -d \
> -p 5000:5000 \
> --name DOCKER-HUB-TARJETA-NOTAS-V1 \
> jezuz1904/to-do-list-v1:1.0
d2df767f546556af856b5d4c012c45fa5baf482b6dab770932fe59af4935030a
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$
```

Comprobamos si los contenedores estas creados



AWS ECR

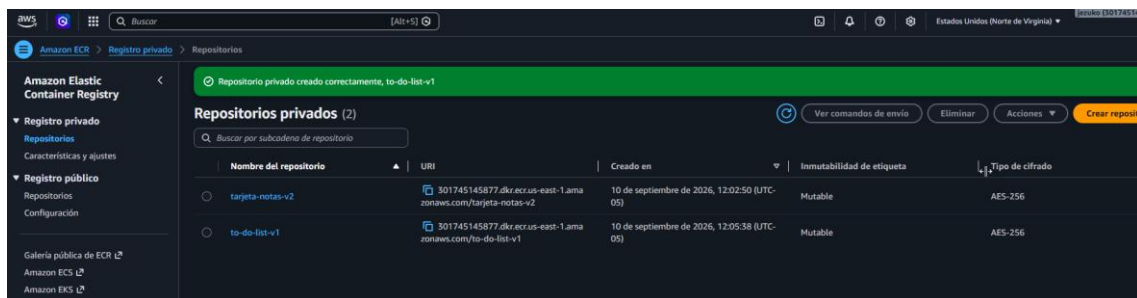
Se replica la misma actividad utilizando AWS ECR. Para esto se requiere una cuenta de AWS previamente creada y configurada; en este caso se utiliza una cuenta con permisos de administrador (jezuz) en la región de Norte de Virginia (us-east-1).

Creación de los repositorios privados en ECR:

Mediante AWS CLI

```
aws ecr create-repository --repository-name tarjeta-notas-v2 --region us-east-1
```

```
aws ecr create-repository --repository-name to-do-list-v1 --region us-east-1
```



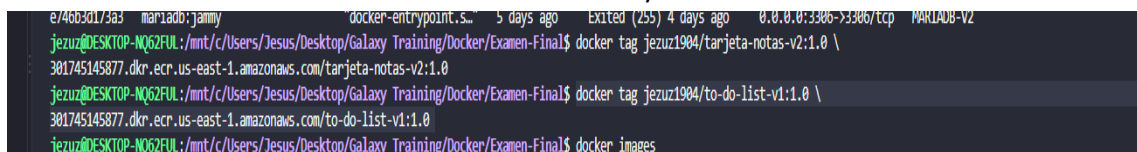
Etiquetado de las imágenes

Antes de publicarlas en ECR, se etiquetan nuevamente con la URL del registro correspondiente a cada repositorio, siguiendo la convención recomendada por AWS:

Tageamos las images

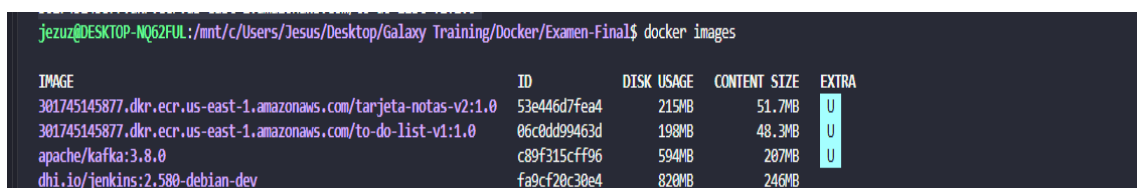
```
docker tag jezuz1904/tarjeta-notas-v2:1.0 \
301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0
```

```
docker tag jezuz1904/to-do-list-v1:1.0 \
301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0
```



Verificación del etiquetado:

```
docker images
```



Publicación de las imágenes en ECR:

docker push 301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0

docker push 301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0

```
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker push 301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0
The push refers to repository [301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2]
8249bd96a5ad: Pushed
44136fa355b3: Pushed
4f6ea04c1543: Pushed
6a0ac1617861: Pushed
b2cbbfe903b0: Pushed
fff4e2c1b189: Pushed
8684042f67ad: Pushed
2b1c10b70b05: Pushed
794b77d4e2cc: Pushed
84fa716861fb: Pushed
49030ba06699: Pushed
1.0: digest: sha256:53e446d7fea4df5d62587ed97734a0ab496b3195764008d6f04cb87d4abce295 size: 856
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker push 301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0
The push refers to repository [301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1]
6310eb16bf42: Pushed
5c5309c50c96: Pushed
44136fa355b3: Pushed
dhd0b7849e6c: Pushed
8de16dab4666: Pushed
1560cfad01dc: Pushed
56235e424563: Pushed
12e7896951c2: Pushed
9dca8d907a9a: Pushed
10cba69f7e13: Pushed
ab7715f487d9: Pushed
1.0: digest: sha256:06c0dd99463d252b282359f7e4da5da1f60c380b7823c15b0685a4aada1212b4 size: 856
```

Revisamos si las imágenes fueron publicadas con éxito

Detalles del repositorio

Nombre del repositorio

tarjeta-notas-v2

URI del repositorio

301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2

Tipo de cifrado

AES-256

Creada a las

10 de septiembre de 2026, 12:02:50 (UTC-05)

Mutabilidad de etiquetas

Mutable

Frecuencia de análisis

Manual

Repositorio ARN

arn:awssecurus-east-1:301745145877:repository/tarjeta-notas-v2

Exclusiones de mutabilidad de etiquetas

-

Detalles del repositorio

Nombre del repositorio

to-do-list-v1

URI del repositorio

301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1

Tipo de cifrado

AES-256

Creada a las

10 de septiembre de 2026, 12:05:38 (UTC-05)

Mutabilidad de etiquetas

Mutable

Frecuencia de análisis

Manual

Descarga de las imágenes publicadas en ECR:

docker pull 301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0

docker pull 301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0

```
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker pull 301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0
1.0: Pulling from tarjeta-notas-v2
Digest: sha256:53e446d7fea4df5d62587ed97734a0ab496b3195764008d6f04cb87d4abce295
Status: Downloaded newer image for 301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0
301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker pull 301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0
1.0: Pulling from to-do-list-v1
Digest: sha256:06c0dd99463d252b282359f7e4da5da1f60c380b7823c15b0685a4aada1212b4
Status: Downloaded newer image for 301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0
301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0
jezuz@DESKTOP-NQ62FUL: /mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0	53e446d7fea4	215MB	51.7MB	U
301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0	06c0dd99463d	198MB	48.3MB	U
apache/kafka3.8.0	c80f315c-ff06	504MB	70.7MB	U

Creación de contenedores a partir de las imágenes descargadas desde ECR:

```
docker container run -d \  
> -p 3000:3000 \  
> --name ECR-TARJETAS-NOTAS-V2 \  
> 301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0  
af671c360d795a29776643e4e9f2def712d074c6826e09f8f2294d04593c3e85
```

```
docker container run -d \  
> -p 5000:5000 \  
> --name ECR-TARJETAS-NOTAS-V1 \  
> 301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0  
4a796b7fda99e5656d5a28a2e166104fb387c9c52af7ab3d59f934bacd42449e
```

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container run -d \  
> -p 3000:3000 \  
> --name ECR-TARJETAS-NOTAS-V2 \  
> 301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0  
af671c360d795a29776643e4e9f2def712d074c6826e09f8f2294d04593c3e85  
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container run -d \  
> -p 5000:5000 \  
> --name ECR-TARJETAS-NOTAS-V1 \  
> 301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0  
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container run -d \  
> -p 5000:5000 \  
> --name ECR-TARJETAS-NOTAS-V1 \  
> 301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0  
4a796b7fda99e5656d5a28a2e166104fb387c9c52af7ab3d59f934bacd42449e
```

Verificación de los contenedores creados:

```
4a796b7fda99e5656d5a28a2e166104fb387c9c52af7ab3d59f934bacd42449e  
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container ps  
CONTAINER ID   IMAGE                                     COMMAND                  CREATED        STATUS        PORTS  
4a796b7fda99   301745145877.dkr.ecr.us-east-1.amazonaws.com/to-do-list-v1:1.0   "gunicorn --bind 0.0.0."  16 seconds ago Up 15 seconds (healthy)  0.0.0.0:5000->5000/tcp, [::]  
af671c360d79   301745145877.dkr.ecr.us-east-1.amazonaws.com/tarjeta-notas-v2:1.0   "docker-entrypoint.s..."  2 minutes ago Up 2 minutes (healthy)  0.0.0.0:3000->3000/tcp, [::]  
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$
```

